

AWS Homelab Security Report (Phase F)

Executive Summary

This report summarizes the design, implementation, testing, and documentation of a secure AWS Homelab environment. A dedicated VPC with public and private subnets was created, CloudTrail and AWS Config were enabled, IAM security controls were configured, and CloudGoat and iam-vulnerable were deployed in isolated private subnets for cybersecurity training.

Security Controls Implemented

Network isolation using VPC and private subnets.
CloudTrail logging enabled.
AWS Config compliance monitoring enabled.
Least-privilege IAM groups and password policy implemented.
CloudGoat deployed in isolated environment.

Findings

The assessment identified intentional vulnerabilities within the lab environment. These included vulnerable IAM configurations and intentionally insecure cloud resources used exclusively for training. No production resources were exposed because all testing occurred inside an isolated laboratory.

Recommendations

Continue enforcing MFA, least privilege, centralized logging, regular IAM reviews, AWS Config compliance rules, and GuardDuty monitoring. Remove vulnerable resources after testing and review CloudTrail logs regularly.

Conclusion

The objectives of the AWS Homelab project were successfully completed. The resulting environment provides a secure foundation for cloud security experimentation while demonstrating AWS security best practices and cloud governance concepts.